BEING A COEQUALIZER IS FPQC LOCAL ON THE TARGET

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted spans carry 1 navigation stub; the step that uses it is written out in the span.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Editorial note. The canonical tag of this label does not occur among the tags of the published items of descent.tex, so it is not a published duplicate.

Allowed standard prerequisites (editorial).

- fpqc coverings are universally effective epimorphisms. Same chapter, Lemma lemma-fpqc-universal-effective-epimorphisms, invoked at proof lines 4448-4449. Supplies that morphisms out of Z are described by compatible families on the covering.
- The universal property of a coequalizer. Standard category theory, used at proof lines 4447-4448. Supplies the property being tested against test schemes.

Source reference. Original lemma 023Q: sources/stacks-descent/pinned-original-source.tex, source lines 4348–4348. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Consider schemes X, Y, Z and morphisms $a, b : X \to Y$ and a morphism $c : Y \to Z$ with $c \circ a = c \circ b$. Set $d = c \circ a = c \circ b$. If there exists an fpqc covering $\{Z_i \to Z\}$ such that*

1. *for all i the morphism $Y \times_{c,Z} Z_i \to Z_i$ is the coequalizer of $(a, 1) : X \times_{d,Z} Z_i \to Y \times_{c,Z} Z_i$ and $(b, 1) : X \times_{d,Z} Z_i \to Y \times_{c,Z} Z_i$, and*
2. *for all i and i' the morphism $Y \times_{c,Z} (Z_i \times_Z Z_{i'}) \to (Z_i \times_Z Z_{i'})$ is the coequalizer of $(a, 1) : X \times_{d,Z} (Z_i \times_Z Z_{i'}) \to Y \times_{c,Z} (Z_i \times_Z Z_{i'})$ and $(b, 1) : X \times_{d,Z} (Z_i \times_Z Z_{i'}) \to Y \times_{c,Z} (Z_i \times_Z Z_{i'})$*

then c is the coequalizer of a and b.

Proof. Namely, for a scheme T a morphism $Z \to T$ is the same thing as a collection of morphism $Z_i \to T$ which agree on overlaps by Lemma 023Q. □

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